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Computing the SSR

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The skew-stickiness-ratio (SSR), examined in detail by Bergomi in his book, is critically important to options traders, especially market makers. We present a model-free expression for the SSR in terms of the characteristic function. In the diffusion setting, it is well-known that the short-term limit of the SSR is 2; a corollary of our results is that this limit is $H + 3/2$ where H is the Hurst exponent of the volatility process. The general formula for the SSR simplifies and becomes particularly tractable in the affine forward variance case. In particular, we explain the qualitative behavior of the SSR with respect to the shape of the forward variance curve.

Keywords: Skew-stickiness ratio; Stochastic volatility; Rough volatility; Forward variance model; Characteristic function; Forest expansion

1. The skew-stickiness-ratio (SSR)

Consider a European call option expiring at T with strike K, valued as of time t. Denote the Black-Scholes implied volatility of this option by $\sigma_{\rm BS,t}(k,T)$ where $k = \log K/S$ is the log-strike, $S = S_t$ the spot. More explicitly, $\sigma_{\rm BS} = \sigma_{\rm BS,t}(k,T)$ is the unique solution of

$$\mathbb{E}_t\left[\left(\frac{S_T}{S_t} - e^k\right)^+\right] = \mathrm{BS}(k,(T-t)\sigma_{\rm BS}^2),$$

where the Black-Scholes call price function is given by

$$\mathrm{BS}(k,w) = \begin{cases} \Phi\left(-\frac{k}{\sqrt{w}} + \frac{\sqrt{w}}{2}\right) & \\ -e^k\,\Phi\left(-\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2}\right) & \text{if } w > 0 \\ (1-e^k)^+ & \text{if } w = 0, \end{cases}$$

and Φ denotes the standard normal distribution function. Equivalently,

$$C_t(K,T) = \mathbb{E}_t\left[(S_T - K)^+\right] = C_{\rm BS,t}(K,T;\sigma_{\rm BS})$$

with $\sigma_{\rm BS} = \sigma_{\rm BS,t}(\log(K/S_t),T)$. Market makers, when updating option prices using the Black-Scholes formula, typically consider two effects. First, the explicit spot effect†

$$\frac{\partial C_{\rm BS,t}}{\partial S_t}\,\delta S_t,$$

and secondly, the change in implied volatility conditional on a change in the spot

$$\frac{\partial C_{\rm BS,t}}{\partial \sigma_{\rm BS}}\,\mathbb{E}_t\left[\delta\sigma_{\rm BS,t}|\delta S_t\right].$$

(The left-hand factor here is just the Black-Scholes vega, evaluated at $\sigma_{\rm BS} = \sigma_{\rm BS,t}(k,T)$.)

ATM implied volatilities $\sigma_t(T) = \sigma_{{\rm BS},t}(0,T)$ and stock prices are both observable. So market makers can estimate the second component for at-the-money (ATM) options using a simple regression:

$$\delta\sigma_t(T) = \beta_t(T)\,\frac{\delta S_t}{S_t} + \text{noise} =: \beta_t(T)\delta X_t + \text{noise}.$$

For a given time to expiration T, we define the ATM volatility skew

$$\mathcal{S}_t(T) = \frac{\partial}{\partial k}\sigma_{\rm BS,t}(k,T)\bigg|_{k=0}.$$

Obviously, $\mathcal{S}_t(T)$ is also a market observable. Bergomi (2009) calls

$$\mathcal{R}_t(T) = \frac{\beta_t(T)}{\mathcal{S}_t(T)}. \tag{1}$$

the *skew-stickiness ratio* or *SSR*.

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† In this introductory discussion, δS_t should be understood as $S_{t+h} - S_t$ for a small but fixed $h > 0$.

This paper is organized as follows. In Section 1, we begin by introducing the concept of the skew-stickiness ratio (SSR) and discuss its relevance in financial modeling. In Section 2, the analysis of the SSR is extended to scenarios with stochastic volatility. In Section 3, the SSR is expressed in terms of the characteristic function, which is the basis of our subsequent analysis.

Section 4 specializes to the case of affine forward variance (AFV) models, with a specific focus on the Classical Heston model. Section 5 details the computation of $R_t(T)$ using the forest expansion method, with step-by-step calculations to second order and leading order. The examples provided illustrate the application of these methods to both the Classical Heston model and the Rough Heston model.

Section 6 explores the dependence of $R_t(T)$ on the forward variance curve, and in Section 7, the sensitivity of the SSR to different model dynamics is discussed. Finally, in Section 8, we summarize our contributions and make some concluding remarks.

2. The SSR under stochastic volatility

Our objective in this paper will be to compute the SSR under stochastic volatility. For concreteness, let's focus on forward variance models of the form

$$\begin{aligned}\frac{\mathrm{d}S_t}{S_t} &= \sqrt{V_t}\,\mathrm{d}Z_t\\ \mathrm{d}\xi_t(u) &= f_t(\xi)\,\kappa(u-t)\,\mathrm{d}W_t,\end{aligned}\tag{2}$$

where $X = \log S$, $V_t\,\mathrm{d}t = \mathrm{d}\,\langle X\rangle_t$, $\xi_t(u) = \mathbb{E}_t\,[V_u]$, and $\mathrm{d}\langle Z, W\rangle_t = \rho\,\mathrm{d}t$. In particular, it follows that $\xi_t(t) = V_t$. Such a model is scale-invariant, with $f_t(\xi) = f_t(\xi_{\bullet\wedge t})$ adapted to the filtration generated by W.

Assuming that we can compute option prices in model (2), computing $\mathcal{S}_t(T)$ is in principle straightforward. If, as is usual in stochastic volatility modeling, we assume that ξ is adapted to the filtration generated by W, the ATM volatility $\sigma_t(T)$ can only depend on the volatility state variables, namely the forward variances $\{\xi_t(u) : u > t\}$. Then, with $X = \log S$,

$$\beta_t(T) = \frac{\frac{\mathrm{d}}{\mathrm{d}t}\langle\sigma(T), X\rangle_t}{\frac{\mathrm{d}}{\mathrm{d}t}\langle X\rangle_t},\tag{3}$$

in terms of the usual quadratic covariation bracket of stochastic analysis. Semimartingality of implied volatility, as seen from time t, with maturity T, as process in t, was discussed by various authors, notably (Durrleman 2010).†

† Strictly speaking, for constant money strike K. The forward ATM case of interest to us, amounts to take $K = S_t$ in which case semimartingality can be obtained via the Itô-Wentzell formula.

Applying Itô's Formula, denoting the functional derivative by δ‡

$$\begin{aligned}\mathrm{d}\,\langle\sigma(T), X\rangle_t &= \int_t^T \mathrm{d}u\,\frac{\delta\sigma_t(T)}{\delta\xi_t(u)}\,\mathrm{d}\,\langle\xi(u), X\rangle_t\\ &= \sqrt{V_t}\int_t^T \mathrm{d}u\,\frac{\delta\sigma_t(T)}{\delta\xi_t(u)}\,\rho\, f_t(\xi)\,\kappa(u-t)\,\mathrm{d}t.\end{aligned}$$

This gives

$$\beta_t(T) = \frac{\rho}{\sqrt{V_t}}\int_t^T \frac{\delta\sigma_t(T)}{\delta\xi_t(u)}\, f_t(\xi)\,\kappa(u-t)\,\mathrm{d}u.\tag{4}$$

In order to streamline notation, let us define the operator

$$D_t^{\xi} := \frac{1}{\sqrt{V_t}}\int_t^T \mathrm{d}u\, f_t(\xi)\,\kappa(u-t)\,\frac{\delta}{\delta\xi_t(u)}.\tag{5}$$

With this notation (4) becomes

$$\beta_t(T) = \rho\, D_t^{\xi}\,\sigma_t(T).\tag{6}$$

REMARK 2.1 The right-hand side of (3) is nothing but equation (9.5) of Bergomi (2015),

$$\beta_t(T) = \frac{\mathbb{E}_t\,[\mathrm{d}\,\langle\sigma(T), X\rangle_t]}{\mathbb{E}_t\,[\mathrm{d}\,\langle X\rangle_t]}.\tag{7}$$

3. The SSR in terms of the characteristic function

Let $\Sigma_t(k,T) = \sigma_{BS,t}(k,T)^2\,(T-t)$. From equation (5.7) of Gatheral (2006) (see also Lewis 2000 and Lipton 2002), we have the following relationship between $\Sigma_t(k,T)$ and the characteristic function $\varphi_t(T;a) = \mathbb{E}_t\left[e^{iaX_T}\right]$

$$\begin{aligned}&\int_{\mathbb{R}^+}\frac{\mathrm{d}a}{a^2+\frac{1}{4}}\,\Re\left[e^{-\mathrm{i}\,a\,k}\,e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right)\Sigma_t(k,T)}\right]\\ &\quad= \int_{\mathbb{R}^+}\frac{\mathrm{d}a}{a^2+\frac{1}{4}}\,\Re\left[e^{-\mathrm{i}\,a\,k}\varphi_t(T;a-\mathrm{i}/2)\right].\end{aligned}\tag{8}$$

Equation (8) leads to the following representation of the SSR $\mathcal{R}_t(T)$ in terms of the characteristic function.

PROPOSITION 3.1 *Let $\psi = \log\varphi$. Then*

$$\mathcal{R}_t(T) = -\frac{\int_{\mathbb{R}^+}\frac{\mathrm{d}a}{a^2+\frac{1}{4}}\,\Re[\rho\, D_t^{\xi}\,\psi_t(T;a-\mathrm{i}/2)\,\exp\{\psi_t(T;a-\mathrm{i}/2)\}]}{\int_{\mathbb{R}^+}\frac{a\,\mathrm{d}a}{a^2+1/4}\,\Im\,[\exp\{\psi_t(T;a-\mathrm{i}/2)\}]}.\tag{9}$$

Proof Differentiating (8) wrt k, and performing the integration on the LHS, we obtain (5.8) of Gatheral (2006):

$$\begin{aligned}\mathcal{S}_t(T) = &-e^{\Sigma_t(0,T)/8}\sqrt{\frac{2}{\pi}}\,\frac{1}{\sqrt{T-t}}\int_{\mathbb{R}^+}\\ &\times\frac{a\,\mathrm{d}a}{a^2+1/4}\,\Im\,[\varphi_t(T;a-\mathrm{i}/2)]\,.\end{aligned}\tag{10}$$

Now we have the skew, we need only compute the regression coefficient $\beta_t(T)$. Evaluating the functional derivative of (8)

‡ See Appendix for a short tutorial on functional derivatives.

with respect to $\xi_t(u)$ at $k = 0$,

$$\int_{\mathbb{R}^+} da\, \Re\left[\frac{\delta \Sigma_t(0,T)}{\delta \xi_t(u)}\, e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right)\Sigma_t(0,T)}\right] = \int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}}\, \Re\left[\frac{\delta}{\delta \xi_t(u)} \varphi_t(T; a - \mathrm{i}/2)\right].$$

The LHS may once again be integrated explicitly to give

$$\frac{\sqrt{2\pi}}{\sqrt{\Sigma_t(0,T)}}\, \frac{\delta \Sigma_t(0,T)}{\delta \xi_t(u)} = e^{\frac{1}{8}\Sigma_t(0,T)} \int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}}\, \Re\left[\frac{\delta}{\delta \xi_t(u)} \varphi_t(T; a - \mathrm{i}/2)\right].$$

Equivalently,

$$\frac{\delta \sigma_t(T)}{\delta \xi_t(u)} = e^{\frac{1}{8}\Sigma_t(0,T)} \sqrt{\frac{2}{\pi}}\, \frac{1}{\sqrt{T-t}} \int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}}\, \Re \times \left[\frac{\delta}{\delta \xi_t(u)} \varphi_t(T; a - \mathrm{i}/2)\right].$$

Substituting into (4), we obtain

$$\begin{aligned}\beta_t(T) &= \frac{\rho}{\sqrt{V_t}} \int_t^T \frac{\delta \sigma_t(T)}{\delta \xi_t(u)} f_t(\xi) \kappa(u-t)\, du \\ &= \rho\, e^{\frac{1}{8}\Sigma_t(0,T)} \sqrt{\frac{2}{\pi}}\, \frac{1}{\sqrt{T-t}} \int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}} \\ &\quad \times \Re\left[D_t^{\xi}\, \varphi_t(T; a - \mathrm{i}/2)\right]. \end{aligned} \tag{11}$$

Substituting from (10) and (11), we obtain

$$\mathcal{R}_t(T) = -\frac{\int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}}\, \rho\, \Re\left[D_t^{\xi} \varphi_t(T; a - \mathrm{i}/2)\right]}{\int_{\mathbb{R}^+} \frac{a\, da}{a^2+1/4}\, \Im\left[\varphi_t(T; a - \mathrm{i}/2)\right]}.$$

With $\varphi = e^{\psi}$, the result follows. ■

4. The SSR in affine forward variance (AFV) models

Recall from Gatheral and Keller-Ressel (2019) that in AFV models,

$$d\xi_t(u) = \kappa(u-t)\sqrt{V_t}\, dW_t, \tag{12}$$

so $f_t(\xi) = \sqrt{V_t}$. The following lemma gives a formula for the SSR in AFV models.

PROPOSITION 4.1 *In an affine forward variance model of the form (12),*

$$\mathcal{R}_t(T) = -\frac{\int_{\mathbb{R}^+} \frac{da}{a^2+\frac{1}{4}}\, \Re\left[\rho\, (\kappa \star g)(T-t; a - \mathrm{i}/2)\, e^{\int_t^T \xi_t(s)\, g(T-s; a-\mathrm{i}/2)\, ds}\right]}{\int_{\mathbb{R}^+} \frac{a\, da}{a^2+1/4}\, \Im\left[e^{\int_t^T \xi_t(s)\, g(T-s; a-\mathrm{i}/2)\, ds}\right]}, \tag{13}$$

where g satisfies the convolution integral Riccati equation

$$\begin{aligned} g(\tau; a) &= -\frac{1}{2}\, a(a+\mathrm{i}) + \mathrm{i}\, \rho\, a\, (\kappa \star g)(\tau; a) \\ &\quad + \frac{1}{2}\, (\kappa \star g)(\tau; a)^2. \end{aligned} \tag{14}$$

Proof From Gatheral and Keller-Ressel (2019),

$$\psi_t(T; a) = \log \varphi_t(T; a) = \int_t^T \xi_t(s)\, g(T-s; a)\, ds, \tag{15}$$

where g satisfies the convolution integral Riccati equation (14). Functionally differentiating (15) gives, for $u \in [t, T]$,

$$\frac{\delta}{\delta \xi_t(u)} \psi_t(T; a) = g(T-u; a).$$

Thus

$$D_t^{\xi}\, \psi_t(T; a - \mathrm{i}/2) = (\kappa \star g)(T-t; a - \mathrm{i}/2).$$

Substitution into (9) yields the result. ■

Given a (numerical or analytical) expression $g(\cdot)$ for the solution of the convolution Riccati equation (14), the expression (13) may be evaluated numerically. In particular, $\mathcal{R}_t(T)$ may be evaluated in the rough Heston model.

4.1. *The classical Heston SSR*

Consider the classical Heston model given by

$$\begin{aligned} \frac{dS_t}{S_t} &= \sqrt{V_t}\, dZ_t \\ dV_t &= -\lambda\, (V_t - \bar{V})\, dt + \nu \sqrt{V_t}\, dW_t, \end{aligned}$$

with $d\langle W, Z\rangle_t = \rho\, dt$. As is well-known (see Gatheral 2006 for example), the characteristic exponent is given by

$$\psi_t(T; a) = \log \mathbb{E}_t\left[e^{\mathrm{i}aX_T}\right] = \mathrm{i}a\, X_t + D(\tau; a)\, V_t + C(\tau; a)\, \bar{V}, \tag{16}$$

where $X = \log S$, $\tau = T - t$, and C and D are complicated functions given in closed-form. The proof of the following lemma shows how the SSR may be computed in a Markovian model.

LEMMA 4.2 *In the classical Heston model,*

$$\mathcal{R}_t(T) = \frac{\int_{\mathbb{R}^+} \frac{da}{a^2+1/4}\, \Re[\rho\nu\, D(\tau; a - \mathrm{i}/2)\, \exp\left\{D(\tau; a-\mathrm{i}/2)\, V_t + C(\tau; a-\mathrm{i}/2)\, \bar{V}\right\}]}{\int_{\mathbb{R}^+} \frac{a\, da}{a^2+1/4}\, \Im[\exp\left\{D(\tau; a-\mathrm{i}/2)\, V_t + C(\tau; a-\mathrm{i}/2)\, \bar{V}\right\}]} \tag{17}$$

Proof The Lewis formula (Lewis 2000), gives the price of a European call option in terms of the characteristic function by

$$\begin{aligned} C_t &= \mathbb{E}_t\left[(S_T-K)^+\right] \\ &= S-\sqrt{SK}\,\frac{1}{\pi}\int_{\mathbb{R}^+}\frac{\mathrm{d}u}{u^2+1/4}\,\Re\left[\varphi_t(T;u-\mathrm{i}/2)\right]. \end{aligned} \tag{18}$$

Following Section 9.8 on page 368 of Bergomi (2015), $\beta_t(T)$ may be computed numerically by shifting the initial instantaneous variance V_t. In infinitesimal terms, and with the notation of Section 1, Bergomi's recipe reads

$$\beta_t(T)=\rho\,\nu\left(\frac{\partial C_{BS,t}}{\partial\sigma_{\rm BS}}\right)^{-1}\frac{\partial C_t}{\partial V_t}.$$

Substituting the explicit form (16) of the characteristic function into (18), and differentiating, we find (with $K=S_t$, taken without loss of generality as equal to 1),

$$\begin{aligned} \frac{\partial C_t}{\partial V_t} &= -\frac{1}{\pi}\int_{\mathbb{R}^+}\frac{\mathrm{d}u}{u^2+1/4}\,\Re \\ &\quad\times\left[D(\tau;a)\,\exp\left\{D(\tau;a)\,V_t+C(\tau;a)\,\bar V\right\}\right]. \end{aligned}$$

Differentiating the Black-Scholes formula, again with $K=S_t=1$, we obtain that

$$\left.\frac{\partial C_{\rm BS,t}}{\partial\sigma_{\rm BS}}\right|_{\sigma_{\rm BS}=\sigma_t(T)}=e^{-\Sigma_t(0,T)/8}\sqrt{\frac{T-t}{2\,\pi}},$$

using the familiar formula for the Black-Scholes Vega. Thus

$$\begin{aligned} \beta_t(T) &= -e^{\Sigma_t(0,T)/8}\rho\,\nu\sqrt{\frac{2}{\pi}}\sqrt{\frac{1}{T-t}} \\ &\quad\times\int_{\mathbb{R}^+}\frac{\mathrm{d}u}{u^2+1/4} \\ &\quad\times\Re\left[D(\tau;a)\,\exp\left\{D(\tau;a)\,V_t+C(\tau;a)\,\bar V\right\}\right]. \end{aligned} \tag{19}$$

Also from (10),

$$\begin{aligned} \mathcal{S}_t(T) &= -e^{\Sigma_t(0,T)/8}\sqrt{\frac{2}{\pi}}\frac{1}{\sqrt{T-t}}\int_{\mathbb{R}^+}\frac{u\,\mathrm{d}u}{u^2+1/4} \\ &\quad\times\Im\left[\exp\left\{D(\tau;a)\,V_t+C(\tau;a)\,\bar V\right\}\right]. \end{aligned} \tag{20}$$

Dividing (19) by (20) yields the result. ■

Since $C(\tau;a)$ and $D(\tau;a)$ have closed-form expressions, $\mathcal{R}_t(T)$ may be computed using straightforward numerical integration. (17) is of course a special case of formula (13); whereas Bergomi's method works only for Markovian models, formula (13) applies to all affine forward variance models.

5. $\mathcal{R}_t(T)$ from the forest expansion

From Alòs *et al.* (2020) and Friz *et al.* (2022), we have the following definition of the diamond product.

Definition 5.1 *Given two continuous semimartingales A, B with integrable covariation process $\langle A,B\rangle$, the diamond product is defined by*

$$(A\diamond B)_t(T):=\mathbb{E}\left[\langle A,B\rangle_{t,T}\middle|\mathcal{F}_t\right]=\mathbb{E}\left[\langle A,B\rangle_T\middle|\mathcal{F}_t\right]-\langle A,B\rangle_t.$$

Diamond products of (sufficiently integrable, continuous) semimartingales naturally lead to binary 'diamond' trees such as $(A\diamond B)\diamond C$. Diagrammatically, the diamond product of two trees $\mathbb{T}_1$ and $\mathbb{T}_2$ is represented by *root joining*,

$$\mathbb{T}_1\diamond\mathbb{T}_2=\ ,$$

where the two binary trees $\mathbb{T}_1$ and $\mathbb{T}_2$ are represented as the single leaves and . We regard linear combinations of diamond trees as *forests*.

From Alòs *et al.* (2020), the cumulant generating function (CGF) is given by the forest expansion

$$\psi_t(T;a):=\log\varphi_t(T;a)=\sum_{\ell=0}^{\infty}\tilde{\mathbb{F}}_\ell(a), \tag{21}$$

where the $\tilde{\mathbb{F}}_\ell$ satisfy the recursion $\tilde{\mathbb{F}}_0=-\frac{1}{2}a(a+\mathrm{i})\,M$ and for $\ell>0$,

$$\tilde{\mathbb{F}}_\ell=\frac{1}{2}\sum_{j=0}^{\ell-2}\left(\tilde{\mathbb{F}}_{\ell-2-j}\diamond\tilde{\mathbb{F}}_j\right)+\mathrm{i}a\left(X\diamond\tilde{\mathbb{F}}_{\ell-1}\right). \tag{22}$$

Applying the recursion (22), the first few $\tilde{\mathbb{F}}$ forests are given by

$$\begin{aligned} \tilde{\mathbb{F}}_0 &= -\tfrac{1}{2}a(a+\mathrm{i}) \\ \tilde{\mathbb{F}}_1 &= -\tfrac{\mathrm{i}}{2}a^2(a+\mathrm{i}) \\ \tilde{\mathbb{F}}_2 &= \tfrac{1}{2^3}a^2\,(a+\mathrm{i})^2 + \tfrac{1}{2}a^3\,(a+\mathrm{i}) \\ \tilde{\mathbb{F}}_3 &= (\tilde{\mathbb{F}}_0\diamond\tilde{\mathbb{F}}_1)+\mathrm{i}a\diamond\tilde{\mathbb{F}}_2 \\ &= \tfrac{\mathrm{i}}{2^2}a^3\,(a+\mathrm{i})^2 + \tfrac{\mathrm{i}}{2^3}a^3\,(a+\mathrm{i})^2 + \tfrac{\mathrm{i}}{2}a^4\,(a+\mathrm{i}) , \end{aligned}$$

where and .

Remark 5.2 Note that the total probability and martingale constraints are satisfied for each tree. That is $\psi_t^T(0)=\psi_t^T(-\mathrm{i})=0$.

5.1. *Computation of $\mathcal{R}_t(T)$ to second order in the forest expansion*

Consider a formal expansion according to values of ℓ, which we term the *forest expansion*. The number of leaves in each tree in the forest $\tilde{\mathbb{F}}_\ell$ is given by $\ell+2$ where counts as one leaf, and $=$ counts as two leaves.

Proposition 5.3 *To second order (i.e. up to $\ell=2$) in the forest expansion,*

$$\mathcal{R}_t(T)=\frac{M_t(T)\,\rho\,D_t^{\xi}\left\{\ +\frac{1}{2}\ \right\}}{\ +\ }. \tag{23}$$

Proof From (21), to second order in the forest expansion,

$$\psi_t(T;a-\mathrm{i}/2) = -\tfrac{1}{2}\left(a^2+\tfrac{1}{4}\right)\left\{\bullet + (\mathrm{i}\,a+\tfrac{1}{2})\,\bullet\!\vee\!\circ - \tfrac{1}{4}\left(a^2+\tfrac{1}{4}\right)\bullet\!\vee\!\bullet - (a-\mathrm{i}/2)^2\,\bullet\!\vee\!\circ\circ\right\}.$$

Then

$$\Re\left[e^{\psi_t(T;a-\mathrm{i}/2)}\,D_t^{\xi}\left\{\bullet + (\mathrm{i}\,a+\tfrac{1}{2})\,\bullet\!\vee\!\circ\right\}\right] \approx \Re\left[e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right)M}\,D_t^{\xi}\left\{\bullet + (\mathrm{i}\,a+\tfrac{1}{2})\,\bullet\!\vee\!\circ\right\}\right]$$
$$= e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right)M}\,D_t^{\xi}\left\{\bullet + \tfrac{1}{2}\,\bullet\!\vee\!\circ\right\},$$

and

$$\Im\left[e^{\psi_t(T;a-\mathrm{i}/2)}\right] = -\frac{1}{2}\,e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right)M}\,a\,\left(a^2+\tfrac{1}{4}\right)\left\{\bullet\!\vee\!\circ + \bullet\!\vee\!\circ\circ\right\}.$$

The result follows by substitution into (9). ∎

5.2. *Computation of $\mathcal{R}_t(T)$ to leading order*

Lemma 5.4 *To leading order,*

$$\mathcal{R}_t(T) = \frac{M_t(T)}{\sqrt{V_t}}\,\frac{\int_t^T f_t(\xi)\kappa(u-t)\,\mathrm{d}u}{\int_t^T \mathrm{d}s\,\int_s^T \mathbb{E}_t\left[\sqrt{V_s}f_s(\xi)\right]\kappa(u-s)\,\mathrm{d}u}. \tag{24}$$

Proof From Proposition 5.3, to leading order,

$$\mathcal{R}_t(T) = \frac{M_t(T)\,\rho\,D_t^{\xi}\bullet}{\bullet\!\vee\!\circ}.$$

Since $\frac{\delta}{\delta\xi_t(u)}M_t(T)=1$, by the definition (5) of D^{ξ},

$$D_t^{\xi}\bullet = \frac{1}{\sqrt{V_t}}\int_t^T f_t(\xi)\,\kappa(u-t)\,du,$$

Also

$$\bullet\!\vee\!\circ = \rho\int_t^T ds\left(\int_s^T \mathbb{E}_t\left[\sqrt{V_s}\,f_s(\xi)\right]\kappa(u-s)\,du\right).$$

∎

The following corollary gives the limit as $T\to t$ of the SSR for any stochastic volatility model of the form (2).

Corollary 5.5 *Let $\tau = T-t$. Then*

$$\lim_{T\to t}\mathcal{R}_t(T) = \tau\,\frac{\mathrm{d}}{\mathrm{d}\tau}\log\left(\int_0^{\tau}\mathrm{d}s\int_0^{s}\kappa(u)\,\mathrm{d}u\right). \tag{25}$$

Proof We have $\xi_t(u) = V_t +$ higherorder and

$$\mathbb{E}_t\left[\sqrt{V_s}f_s(\xi)\right] = \sqrt{V_t}f_t(\xi) + \text{ higherorder}.$$

Thus to leading order as $T\to t$, $M_t(T)\sim V_t\,\tau$ and from (24),

$$\mathcal{R}_t(T) = \frac{M_t(T)}{V_t}\,\frac{\int_t^T\kappa(u-t)\,\mathrm{d}u}{\int_t^T\mathrm{d}s\,\left(\int_s^T\kappa(u-s)\,\mathrm{d}u\right)}$$
$$= \tau\,\frac{\mathrm{d}}{\mathrm{d}\tau}\log\left(\int_0^{\tau}\mathrm{d}s\int_0^{s}\kappa(u)\,\mathrm{d}u\right).$$

∎

The following corollary is then in agreement with formal computation of Fukasawa in Remark 2.10 of Fukasawa (2021).

Corollary 5.6 *Let $\kappa(s)\sim s^{\alpha-1}$ as $s\to 0$. Then*

$$\lim_{t\uparrow T}\mathcal{R}_t(T) = \alpha+1.$$

Remark 5.7 In terms of the Hurst parameter $H>0$, one has $\alpha\equiv H+1/2$, so that in the $t\uparrow T$ limit, $\mathcal{R}_t(T)\sim 3/2+H$, confirming a formal computation of Fukasawa in Remark 2.10 of Fukasawa (2021). In particular, one has, when $T\to t$ and in diffusive situations with $H=2$, skew-stickiness-ratio 2, cf. Chapter 9 in Bergomi (2015).

These (asymptotic) results are related to various 'rules' that relate local and implied volatility skews. In a diffusive setting the rule that the local volatility skew is double the implied volatility skew has been known for a long time (Gatheral 2006); this was recently extended to a $H+3/2$-rule in the rough volatility regime $H<1/2$, using a combination of Malliavin calculus and rough analysis (Bourgey *et al.* 2023) (see also Alòs *et al.* 2023). To appreciate this relation, it suffices to note that asymptotically local variance is the expectation of instantaneous variance conditional on a small move in the underlying (e.g. De Marco and Friz 2018 and references therein) and that the short term limit of implied variance is instantaneous variance (Durrleman 2010)—precisely consistent with the definition of $\beta_t(T)$.

5.3. *Computation of $\mathcal{R}_t(T)$ to next-to-leading order in τ*

The forest expansion (21) is effectively a small ν (vol-of-vol) expansion. As noted in Chapter 9 of Bayer *et al.* (2023), with $\tau - T - t$ and for fixed a, the $\tilde{\mathbb{F}}_{\ell}(\tau)$ scale as $\tau^{\ell\alpha+1}$ as $\tau\downarrow 0$. (Recall $\alpha = H+1/2$ so that $\alpha\in(1/2,1)$ in the rough volatility regime under consideration here.) When computing $\mathcal{R}_t(T)$ using (9), different powers of a will, after integration, generate different powers of τ. Thus, to get a small τ expansion of the SSR, it is not sufficient to go to second order in the forest expansion; we also need to take powers of a into account.

$a^n\,\mathbb{F}^{\ell}$, when integrated over the positive real line with respect to the Gaussian kernel $e^{-\frac{1}{2}(a^2+\frac{1}{4})M}$, scales as $\tau^{-(n+1)/2}\tau^{\ell\alpha+1} = \tau^{\ell\alpha-(n-1)/2}$. Thus, we get the following table of examples of relative scaling of typical terms (dividing by $\sqrt{\tau}$):

$\bullet$	1
$\bullet\!\vee\!\circ$	τ^{α}
$\bullet\!\vee\!\circ\circ$	$\tau^{2\alpha}$
$a^2\,\bullet\!\vee\!\bullet$	$\tau^{2\alpha-1}$
$a^2\,\bullet\!\vee\!\bullet\circ$	$\tau^{2\alpha-1}$
$a^2\,(\bullet\!\vee\!\circ)^2$	$\tau^{2\alpha}$
$a^2\,\bullet\,\bullet\!\vee\!\circ$	τ^{α}
$a^4\,(\bullet\!\vee\!\circ)^2$	$\tau^{2\alpha-1}$
$a^6\,\bullet\!\vee\!\bullet^3$	$\tau^{3\alpha-1}$

Taking the above scaling carefully into account yields the following proposition.

Proposition 5.8 *To next-to-leading order in $\tau = T - t$,*

$$\mathcal{R}_t(T) = \rho\, M \left\{ \frac{D_t^\xi + \frac{1}{2} D_t^\xi - \frac{1}{4\,M} D_t^\xi - \frac{1}{M} D_t^\xi - \frac{1}{4\,M} D_t^\xi + \frac{3}{2\,M^2} D_t^\xi + \frac{3}{8M^2} D^\xi + \frac{3}{2M^2} D^\xi - \frac{15}{8M^3} D^\xi{}^2 + \ldots}{+ - \frac{3}{4\,M} (\;)^2 - \frac{105}{24M^3}{}^3 + \frac{15}{8M^2} + \frac{15}{2M^2} - \frac{3}{4M} - \frac{3}{2M} - \frac{3}{M} + \ldots} \right\}, \tag{26}$$

where $+\cdots$ denotes terms $o(\tau^{2\alpha-1})$ upstairs and $o(\tau^{3\alpha})$ downstairs.

Proof To consistently expand to next-to-leading order in τ, the highest order forest that can contribute is $\tilde{\mathbb{F}}_3$. From (21), to third order in the forest expansion,

$$\psi_t(T; a - \mathrm{i}/2) = -\tfrac{1}{2}\left(a^2 + \tfrac{1}{4}\right)\Big\{ + (\mathrm{i}\, a + \tfrac{1}{2}) - \tfrac{1}{4}\left(a^2 + \tfrac{1}{4}\right) - (a - \mathrm{i}/2)^2$$
$$- \tfrac{\mathrm{i}}{2}(a^2 + \tfrac{1}{4})(a - \mathrm{i}/2) - \tfrac{\mathrm{i}}{4}\left(a^2 + \tfrac{1}{4}\right)(a - \mathrm{i}/2) - \mathrm{i}\,(a - \mathrm{i}/2)^3 \Big\}.$$

For the numerator, to next-to-leading order in τ, we obtain

$$\Re\left[e^{\psi_t(T;a-\mathrm{i}/2)}\, D_t^\xi \psi_t(T; a - \mathrm{i}/2)\right] = -\frac{1}{2}\left(a^2 + \frac{1}{4}\right) e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right) M} \Big\{$$
$$D_t^\xi + \tfrac{1}{2} D_t^\xi - \frac{a^2}{4} D^\xi - a^2 D^\xi - \frac{1}{4} a^2 D^\xi$$
$$+ \frac{1}{2} a^4 D^\xi + \frac{1}{8} a^4 D^\xi + \frac{1}{2} a^4 D^\xi - \frac{1}{8} a^6 D^\xi{}^2 + \ldots \Big\}.$$

Then we do the same for the denominator, including terms up to $\tau^{2\alpha+1}$ or $\tau^{3\,\alpha}$.

$$\Im\left[e^{\psi_t(T;a-\mathrm{i}/2)}\right] = -\frac{1}{2} a \left(a^2 + \frac{1}{4}\right) e^{-\frac{1}{2}\left(a^2+\frac{1}{4}\right) M} \Big\{$$
$$+ - \tfrac{1}{4}\, a^2\, (\;)^2 - \frac{1}{24} a^6{}^3 + \frac{1}{8} a^4 + \frac{1}{2} a^4 - \frac{a^2}{4} - \frac{a^2}{2} - a^2 + \ldots \Big\}.$$

Performing the integrations in (9) explicitly yields the result. ∎

Before proceeding with examples of the application of Proposition 5.8, we present a useful lemma that applies only to AFV models, and that incidentally justifies our choice of definition of the operator D^ξ.

Lemma 5.9 *Let $\mathbb{T}$ be a diamond tree in an affine forward variance (AFV) model. Then*

$$(\mathbb{T} \diamond X)_t(T) = \rho \int_t^T \xi_t(s)\, D_s^\xi \mathbb{T}_s(T)\, \mathrm{d}s.$$

Proof From Lemma 4.3 of Friz *et al.* (2022), all diamond trees in an AFV model take the form

$$\mathbb{T}_t(T) = \int_t^T \xi_t(u)\, h(T - u)\, \mathrm{d}u,$$

for some integrable function h. Then by the definition (5) of D_t^ξ,

$$D_t^\xi \mathbb{T}_t(T) = \int_t^T \kappa(u - t)\, h(T - u)\, \mathrm{d}u.$$

On the other hand,

$$(\mathbb{T} \diamond X)_t(T) = \rho \int_t^T \xi_t(s)\, \mathrm{d}s \int_s^T \kappa(u - s)\, h(T - u)\, \mathrm{d}u.$$

∎

Remark 5.10 It is straightforward to check, for example by examining explicit computations under rough Bergomi in Chapter 9 of Bayer *et al.* (2023), that Lemma 5.9 does not hold in general for models that are not affine.

5.3.1. Example: classical Heston with $\lambda = 0$ and $\xi_t(u) = V$. In this (probably simplest possible) case, we have

$$M = \; = V\,\tau$$
$$= \tfrac{1}{2}\,\rho\,\nu\,V\,\tau^2$$
$$= \tfrac{1}{3}\,\nu^2\,V\,\tau^3$$
$$= \tfrac{1}{6}\,\rho^2\,\nu^2\,V\,\tau^3$$
$$= \tfrac{1}{8}\,\rho\,\nu^3\,V\,\tau^4$$
$$= \tfrac{1}{12}\,\rho\,\nu^3\,V\,\tau^4$$
$$= \tfrac{1}{24}\,\rho^3\,\nu^3\,V\,\tau^4.$$

Also, since the forward variance curve is flat and the model is affine, $D^\xi \equiv \nu\,\partial_V$ so

$$D^\xi = \nu\,\tau$$
$$D^\xi = \tfrac{1}{2}\,\rho\,\nu^2\,\tau^2$$
$$D^\xi = \tfrac{1}{3}\,\nu^3\,\tau^3$$
$$D^\xi = \tfrac{1}{6}\,\rho^2\,\nu^3\,\tau^3.$$

Substituting into (26), we obtain

$$\mathcal{R}_t(T) = 2\,\frac{1 + \frac{1}{8}\,\nu\rho\tau + \frac{1}{24}\,\frac{\nu^2\tau}{V} - \frac{1}{96}\,\frac{\rho^2\nu^2\tau}{V} + \mathcal{O}(\tau^2)}{1 - \frac{1}{24}\rho\,\nu\tau + \frac{1}{8}\,\frac{\nu^2\tau}{V} - \frac{3}{32}\,\frac{\rho^2\nu^2\tau}{V} + \mathcal{O}(\tau^2)},$$

which agrees exactly with the exact expansion of the closed-form classical solution.

5.3.2. Example: rough Heston with $\lambda = 0$ and $\xi_t(u) = V$. In the rough Heston case, we don't have a closed-form expression for the characteristic function but we do have rational approximations (Gatheral and Radoičić 2019, 2024). The diamond trees required to implement (26) are straightforward to compute and are given by (see page 193 of Bayer *et al.* 2023 for example):

$$M = \; = V\,\tau$$
$$= \frac{\rho\,\nu}{\Gamma(2+\alpha)}\,V\,\tau^{\alpha+1}$$
$$= \frac{\nu^2}{\Gamma(1+\alpha)^2}\,V\,\frac{\tau^{2\alpha+1}}{2\alpha+1}$$
$$= \frac{\rho^2\nu^2}{\Gamma(2+2\alpha)}\,V\,\tau^{2\alpha+1}$$
$$= \frac{\rho\,\nu^3}{\Gamma(1+\alpha)\,\Gamma(1+2\alpha)}\,V\,\frac{\tau^{3\alpha+1}}{3\alpha+1}$$
$$= \frac{\rho\,\nu^3\,\Gamma(1+2\alpha)}{\Gamma(1+\alpha)^2\Gamma(2+3\alpha)}\,V\,\tau^{3\alpha+1}$$
$$= \frac{\rho^3\,\nu^3}{\Gamma(2+3\alpha)}\,V\,\tau^{3\alpha+1}.$$

Following the notation of Bayer *et al.* (2023), let

$$I_t^{(j)}(T) = \int_t^T \xi_t(s)\,(T-s)^{j\alpha}\,\mathrm{d}s.$$

Then from the definition (5) of D_t^{ξ}, with $\kappa(\tau) = \frac{\nu\tau^{\alpha-1}}{\Gamma(\alpha)}$,

$$\begin{aligned} D_t^{\xi} I_t^{(j)}(T) &= \frac{1}{\sqrt{V_t}} \int_t^T \mathrm{d}u\, f_t(\xi)\kappa(u-t)\, \frac{\delta}{\delta\xi_t(u)} I_t^{(j)}(T) \\ &= \frac{\nu}{\Gamma(\alpha)} \int_t^T (u-t)^{\alpha-1}\,(T-u)^{j\alpha}\,\mathrm{d}u \\ &= \frac{\Gamma(1+j\alpha)}{\Gamma(1+(j+1)\alpha)}\,\tau^{(j+1)\alpha}. \end{aligned}$$

It follows that

$$\begin{aligned} D^{\xi} &= \frac{\nu}{\Gamma(1+\alpha)}\,\tau^{\alpha} \\ D^{\xi} &= \frac{\rho\,\nu^2}{\Gamma(1+2\alpha)}\,\tau^{2\alpha} \\ D^{\xi} &= \frac{\nu^3}{\Gamma(1+\alpha)^2}\,\frac{\Gamma(1+2\alpha)}{\Gamma(1+3\alpha)}\,\tau^{3\alpha} \\ D^{\xi} &= \frac{\rho^2\nu^3}{\Gamma(1+3\alpha)}\,\tau^{3\alpha}. \end{aligned}$$

REMARK 5.11 Note the general pattern, true only for AFV models when the forward variance curve is flat, whereby for a given tree $\mathbb{T}$, $\rho\, V_t\, D_t^{\xi}\mathbb{T}_t(T) = \partial_\tau\,(\mathbb{T} \diamond X)_t(T)$. This is a straightforward corollary of Lemma 5.9.

In figure 1, with $\xi = 0.025$, $\nu = 0.4$, and $\rho = -.8$, we plot the SSR for various values of H, numerically using (13) and using the forest formula (26) with the above substitutions. We observe clear consistency of the forest formula with the numerical computations. However, in the practically relevant case of $H \lessapprox 0.2$, the timescale over which these two agree is arguably too short to be of any practical interest.

6. Dependence of $\mathcal{R}_t(T)$ on $\xi_t(u)$

We may rewrite the leading order expression (24) suggestively in the form

$$\mathcal{R}_t(T) = \frac{\left(\int_t^T \xi_t(s)\,\mathrm{d}s\right) \int_t^T \sqrt{V_t} f_t(\xi)\kappa(u-t)\,\mathrm{d}u}{V_t \int_t^T \mathrm{d}s \int_s^T \mathbb{E}_t\left[\sqrt{V_s} f_s(\xi)\right]\kappa(u-s)\,\mathrm{d}u}. \qquad (27)$$

We first observe that (27) should be rather insensitive to the level of the forward variance curve. On the other hand, $\mathcal{R}_t(T)$ is clearly sensitive to the shape of $\xi_t(u)$. A monotonic increasing forward variance curve will cause the SSR to increase relative to the flat curve case, and vice versa.

In the AFV case, (27) simplifies to

$$\mathcal{R}_t(T) = \frac{\left(\int_t^T \xi_t(s)\,\mathrm{d}s\right)\,\tilde{\kappa}(T-t)}{\int_t^T \xi_t(s)\,\tilde{\kappa}(T-s)\,\mathrm{d}s},$$

where $\tilde{\kappa}(\tau) = \int_0^\tau \kappa(s)\,\mathrm{d}s$. If the forward variance curve is flat with $\xi_t(u) = \bar{V}$, the SSR does not depend on the level V at all! However, once again, $\mathcal{R}_t(T)$ does depend on the shape of $\xi_t(u)$.

In figure 2, with parameters $H = 0.1$; $\nu = 0.4$; $\rho = -.8$ we plot the rough Heston SSR using the AFV formula (13), and assuming three different forward variance curves of the form:

$$\xi_t(u) = (V_t - \bar{V})\,e^{-\lambda t} + \bar{V},$$

with $\bar{V} = 0.025$, $\lambda = 7$ and $V_t = 0.025$ (flat), $V_t = 0.005$ (contango), and $V_t = 0.045$ (backwardation). We see that in fact, the SSR is very sensitive to the shape of the forward variance curve.

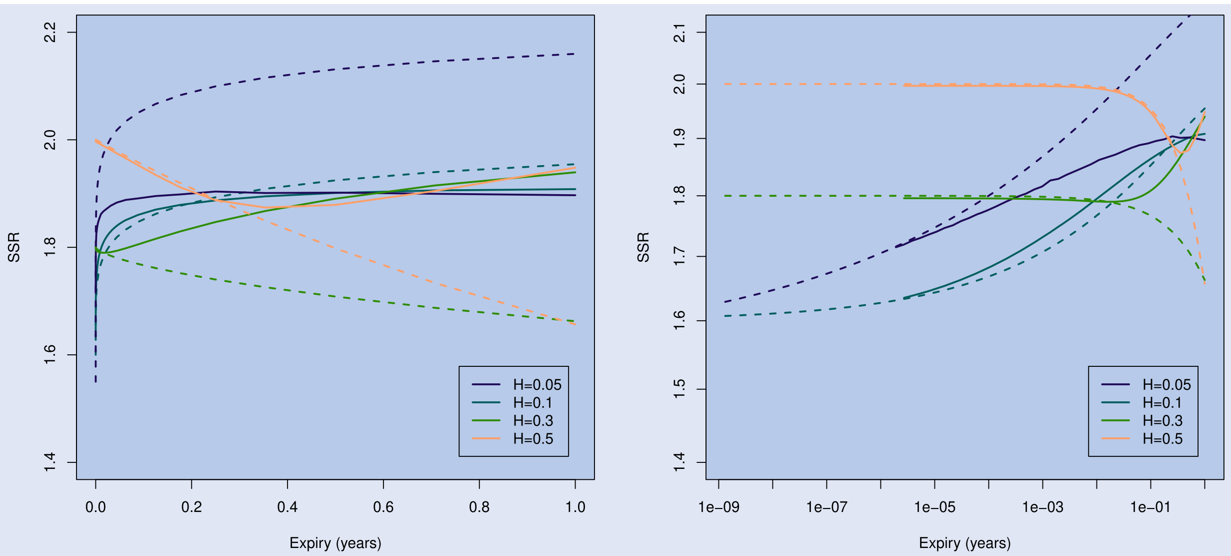

Figure 1. The rough Heston SSR with a flat forward variance curve $\xi = 0.025$, and parameters $\nu = 0.4$; $\rho = -.8$. Solid lines are the numerical computation (13) and dashed lines are the forest formula (26). On the right is a log-log plot.

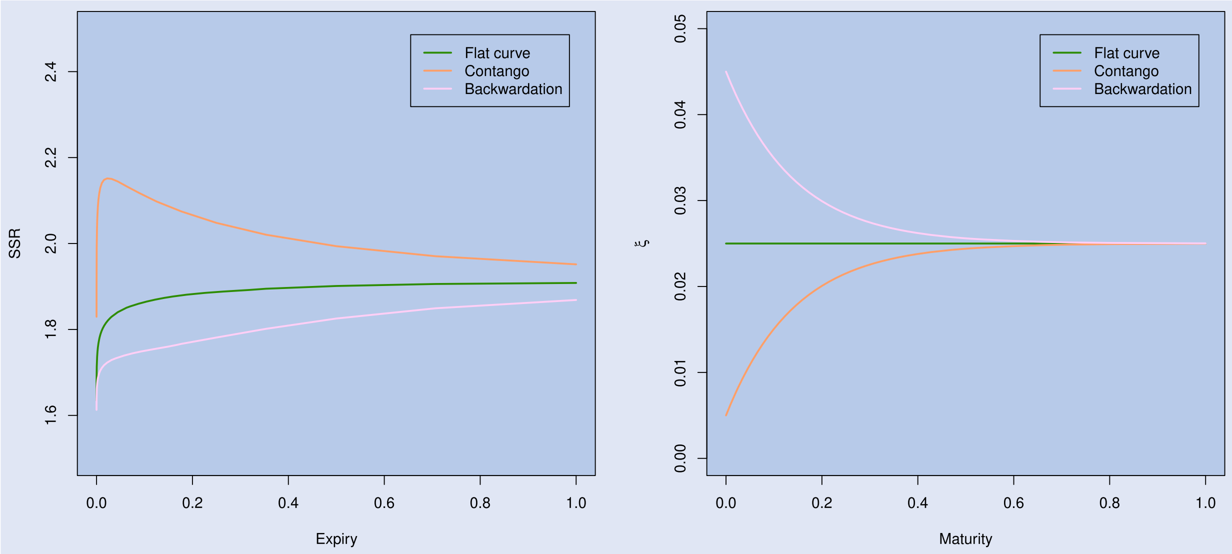

Figure 2. The rough Heston SSR with various forward variance curves, and parameters $H = 0.1$; $\nu = 0.4$; $\rho = -.8$. The left-hand plot is of the SSRs and the right-hand plot shows the assumed forward variance curves.

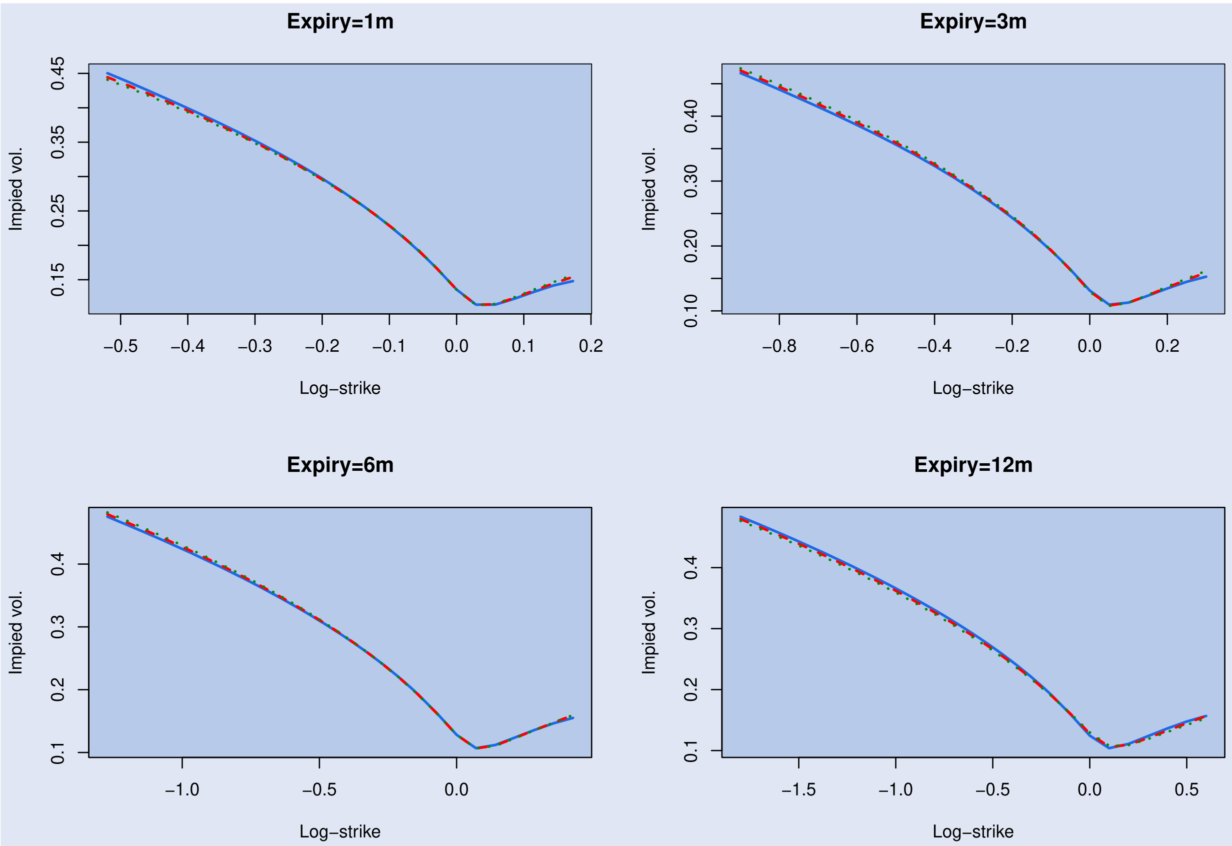

Figure 3. Almost identical rough Heston smiles generated by the parameter sets Π_0, Π_1, and Π_2, in blue, red and green respectively.

Table 1. Parameter values for the different parameter sets.

	Π_0	Π_1	Π_2
λ	0	1	2
H	0.10	0.223	0.302
ν	0.40	0.481	0.647
ρ	-0.65	-0.65	-0.65

7. Sensitivity of the SSR to model dynamics

Recall from Gatheral and Keller-Ressel (2019) that the rough Heston kernel takes the form

$$\kappa(\tau) = \nu \tau^{\alpha-1} E_{\alpha,\alpha}(-\lambda \tau^{\alpha}),$$

where $E_{\alpha,\alpha}(\cdot)$ is the Mittag-Leffler function and $\alpha = H + \frac{1}{2}$. In order to demonstrate the sensitivity of the SSR to model dynamics, we choose three values of the parameter λ (=

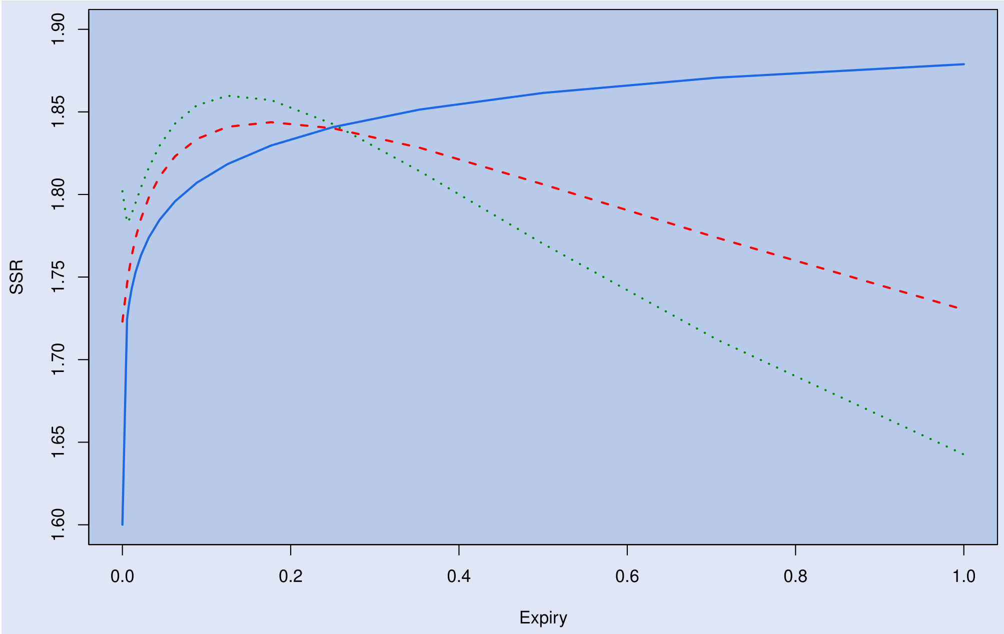

Figure 4. The blue, red, and green SSR plots correspond to parameter sets Π_0, Π_1, and Π_2 respectively.

0, 1, 2) and find values of H and ν such that the resulting parameter sets Π_0, Π_1 and Π_2 (listed in table 1) generate the almost identical 1-month, 3-month, 6-month, and 12-month smiles shown in figure 3.

In figure 4, we plot the SSR, computed using the affine SSR formula (13), for each of these parameter sets as a function of time to expiry. We see that although all three parameter sets generate very similar smiles, the SSR plots are very different. We conclude that the SSR is highly sensitive to dynamical assumptions; it cannot be deduced from the shape of the implied volatility surface.

8. Concluding remarks

To conclude, the key contributions of this paper are twofold. Firstly, Proposition 3.1 gives a generic and essentially model-free expression for the SSR in terms of the characteristic function. Secondly, Proposition 4.1 gives an explicit formula for the SSR in affine forward variance (AFV) models, in terms of the solution of a convolution Riccati integral equation.

The forest expansion (Alòs *et al.* 2020, Friz *et al.* 2022) (closely related to the Bergomi-Guyon expansion (Bergomi and Guyon 2012)) provides a generic way to approximate the characteristic function of a random variable. It is natural to employ this methodology in the context of our Proposition 3.1 (that is also formulated in great generality). From detailed computations and numerical experiments, we conclude that though the forest expansion is theoretically useful (e.g. in deriving short dated asymptotics), it is otherwise of rather limited practical value.

It is important to highlight that observed skew stickiness ratios are typically between 1.0 and 1.3, seemingly inconsistent with our computations in affine models as illustrated in figure 4. Similarly, simulation results of Bourgey *et al.* (2024) suggest that that observed SSRs are inconsistent with the rough Bergomi model. However, we have also established that the SSR is highly dependent on assumed model dynamics, and this gives us hope that a model may be found that generates SSRs consistent with observation.

Indeed, such a model appeared at the revision stage of our paper: The Quintic model, explored in detail in Li (2024), appears to have the desired properties. Underlying the analysis is an expansion of the characteristic function (Abi Jaber *et al.* 2024) and a version of our Proposition 3.1. This also gives evidence that Proposition 3.1 is practically applicable to a much wider class of models than merely those with closed-form characteristic functions or affine models.

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Appendix

A.1. Functional derivatives

We give a short informal tutorial on functional derivatives, focusing on their application in our paper. Fix $t \in [0,T]$ and consider continuous curves $(\xi_t(u) : u \in [t,T])$. We discuss functional calculus on $[t,T]$. Assume the integral

$$I(\xi_t) = \int_t^T \xi_t(u)\, g(u)\, \mathrm{d}u$$

exists. We call $\delta I/\delta\xi_t = g$ the functional derivative, we also write $\delta I/\delta\xi_t(u) = g(u)$ to be understood in the sense of a.s. identity in $u \in [t,T]$. The functional derivative extends to cylindrical functionals via the chain-rule. For $\phi \in C^1$ and $I_i(\xi_t) = \int_t^T \xi_t(u) g_i(u) \mathrm{d}u$,

$$J(\xi_t) = \phi(I_1(\xi_t),\ldots,I_n(\xi_t))$$
$$\Rightarrow \frac{\delta J(\xi_t)}{\delta \xi_t} = \sum_i \partial_i \phi(I_1(\xi_t),\ldots,I_n(\xi_t)) \times g_i.$$

Consider now a family $\{I_a(.) : a \in \mathbb{R}\}$, defined as before with g_a replacing g_i. Assuming suitable integrability properties, Fubini shows that

$$\bar{I}(\xi_t) := \int I_a(\xi_t) p(a)\, \mathrm{d}a = \int_t^T \xi_t(u) \left(\int g_a(u) p(a)\, \mathrm{d}a \right) \mathrm{d}u$$
$$\Rightarrow \frac{\delta \bar{I}(\xi_t)}{\delta \xi_t} = \int g_a(.) p(a)\, \mathrm{d}a =: \bar{g}$$

This again extends to cylindrical functions, $\bar{J}(\xi_t) = \phi(\bar{I}_1(\xi_t),\ldots,\bar{I}_n(\xi_t))$ and so on, leading to a large class $\mathcal{C} = \mathcal{C}_t$ of cylindrical functionals where functional derivatives can be algebraically defined. By (15), the characteristic function of affine forward variance models falls in this class, $\varphi_t(T,a) \in \mathcal{C}$.

Consider now a family $\{J_a(.) : a \in \mathbb{R}\}$, $J_a \in \mathcal{C}$, with nice enough weights $p(.)$ so that

$$\bar{J}(\xi_t) := \int J_a(\xi_t) p(a)\, \mathrm{d}a$$

exists (e.g. as $\lim_{h\downarrow 0} \sum_{a \in h\mathbb{Z}, h|a| \leqslant 1} J_a(\xi_t) p(a) \times h$). In general $\bar{J} \notin \mathcal{C}$, but we can expect (in good situations)

$$\frac{\delta \bar{J}}{\delta \xi_t(u)} = \int \frac{\delta J_a(\xi_t)}{\delta \xi_t(u)} p(a)\, \mathrm{d}a,$$

as may be checked by hand via the approximating sums, so that $\bar{J} \in \bar{\mathcal{C}}_t$, a superset of $\mathcal{C} = \mathcal{C}_t$ on which the derivative is still defined. Let us note that the above discussion is sufficient to 'functionally differentiate' all objects of interest to us, as in (the proof of) Proposition 4.1 as long as the characteristic function $\varphi_t \in \overline{\mathcal{C}_t}$. (Recall that $\varphi_t \in \mathcal{C}_t$ in the affine forward variance case, so this is a very reasonable assumption.)

A.2. A covariation formula

Now consider the dynamics in t. First note that $I(\xi_t) = \int_t^T \xi_t(u) g(u)\, \mathrm{d}u$ is a semimartingale, at least under some uniformity assumption not discussed here.

Consider a (continuous) semimartingale $(\Xi_t : t \in [0,T])$ of the form

$$\Xi_t(T) = J(\xi_t) = \phi(I_1(\xi_t),\ldots,I_n(\xi_t))$$

(In particular, $\Xi_t(T) \in \mathcal{C}_t$ for all $t \in [0,T]$.) By Itô's formula,

$$\begin{aligned} \mathrm{d}_t \Xi_t(T) &= \sum_i \partial_i \phi(I_1(\xi_t),\ldots,I_n(\xi_t)) \times \mathrm{d}_t I_i(\xi_t) + \mathrm{BV} \\ &= \int_t^T \mathrm{d}u \left(\sum_i \partial_i \phi(I_1(\xi_t),\ldots,I_n(\xi_t)) g_i(u) \right) \mathrm{d}_t \xi_t(u) + \mathrm{BV} \\ &= \int_t^T \mathrm{d}u \frac{\delta J(\xi_t)}{\delta \xi_t(u)}\, \mathrm{d}_t \xi_t(u) + \mathrm{BV}, \end{aligned}$$

where BV denotes terms of bounded variation. If X denotes another (continuous) semimartingale, then

$$\mathrm{d}\langle \Xi_t(T), X\rangle_t = \int_t^T \frac{\delta J(\xi_t)}{\delta \xi_t(u)}\, \mathrm{d}\langle \xi(u), X\rangle_t = \int_t^T \frac{\delta \Xi_t(T)}{\delta \xi_t(u)}\, \mathrm{d}\langle \xi(u), X\rangle_t.$$

(A rigorous justification here would again pass over approximations with sums.)